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The Impact of Differentiated Post-Consumer Textile Waste Sorting on the Homogeneity of Recycled Yarns

Magdalena Kohler
Universität der Künste Berlin, Germany

Holger Cebulla
Technical University Chemnitz, Germany

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The Impact of Differentiated Sorting and Mechanical Recycling Processes on Nep Formation and Yarn Quality from Post-Consumer Textil Waste

Kohler, Magdalena*[a]; Cebulla, Holger[a];

[a] Chemnitz University of Technology, Chemnitz, Germany
* m.kohler@udk-berlin.de

Post-consumer textile waste (PCTW) is increasingly characterized by complex multi-fibre blends, posing significant challenges for fibre-to-fibre recycling. This study investigates the influence of sorting, cutting, and tearing processes on nep formation and the resulting yarn quality during mechanical recycling. A three-stage sorting approach based on fibre composition and stitch density was developed and combined with systematically varied cutting and tearing processes. The results show that both input material characteristics and recycling parameters affect nep formation. In particular, thermally melted cut edges from garments containing polyacrylonitrile were identified as a potential source of neps.

The findings highlight the importance of differentiated sorting and process optimization for improving recycled fibre quality and supporting high-quality fibre-to-fibre recycling of PCTW.

Keywords: Post-Consumer Textile Waste; Mechanical Recycling; Sorting; Nep Formation;

1. Introduction

The use of synthetic fibres in textiles offers both economic and functional advantages, as they reduce production costs and enhance the performance of natural fibres when applied in fibre blends (Forst, 2018). Petroleum-based synthetic fibres such as polyacrylonitrile (PAN), polyester (PES), and polyamide (PA) are widely blended with natural fibres to improve durability, dimensional stability, moisture management, and wearer comfort (Hofer, 1981). While early fibre blends primarily consisted

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of two fibre types, textile products have evolved over the past decades towards increasingly complex multi-fibre compositions. In many textiles, natural and synthetic fibres are combined in different blend ratios, with natural fibres often present only in small proportions to maintain a nominal natural content while contributing only marginally to the overall textile performance. This trend is particularly seen in the fast-fashion industry, where garments contain blends of up to seven different fiber types. These developments have occurred largely without consideration of end-of-life management, and recycling remains insufficiently integrated into textile product design.

The consequences of this development have led to rapidly increasing volumes of textile waste. According to the European Parliament, the average European purchases approximately 26 kilograms of textiles per year, of which around 11 kilograms are discarded. An estimated 87% of this waste is either landfilled or incinerated (Europäisches Parlament, 2024). A report by Greenpeace from 2023 indicates that global clothing production doubled between 2000 and 2014, with individuals purchasing 60% more garments annually, despite the fact that these items are worn only half as long. (Cobbing et al., n.d.) In 2014, over 100 billion garments were produced; projections suggest this figure will rise to more than 200 billion by 2030. The Ellen MacArthur Foundation estimates that by then, 134 million tons of textiles will be discarded each year (Ellen Mac Arthur Foundation, 2021).

In contrast to pre-consumer and post-industrial waste streams, post-consumer textile waste (PCTW) exhibits substantial variability in fibre composition and textile construction, making it particularly challenging to recycle. Consequently, effective recycling technologies capable of processing heterogeneous textile waste streams are becoming increasingly important.

Mechanical recycling represents one of the most established recycling technologies for textile fibre recovery (European Commission. Directorate General for Internal Market, Industry, Entrepreneurship and SMEs., 2021; Stefan, 2025). The process typically involves fabric cutting, tearing (shredding), carding, and subsequent spinning to convert textile waste into new yarns.

One of the major challenges in mechanical recycling is fibre length reduction, which significantly impairs yarn spinnability. Shorter fibers exhibit reduced inter-fiber cohesion and are more difficult to control during drafting and spinning, leading to increased fiber loss, higher yarn irregularity, and lower yarn strength (Baloyi et al., 2024; Kohler & Cebulla, 2026; Theobald, 2005). While fibre length reduction is widely recognized as a primary quality limitation, nep formation represents an equally important impact on the performance of recycled fibres (Kugler & Liebhold, 2023). Neps are localized fiber agglomerations consisting of entangled or tightly packed fibers. Neps may also originate from impurities, such as seed coat fragments in cotton or other contaminating particles incorporated during processing. The impact of thermally melted cut edges generated during the cutting of synthetic textiles on subsequent nep formation remains unexplored. Excessive nep content result in processing difficulties, reduced yarn quality, and visible defects in the final textile product.

Despite the importance of nep formation for recycled fibre quality, limited research has addressed the contribution of individual recycling stages and textile construction parameters to nep generation. This study therefore investigates the influence of mechanical recycling processes on nep formation and the resulting yarn quality. To support high-quality fibre-to-fibre recycling of PCTW, a three-stage sorting approach was developed that considers both fibre composition and the stitch density of knitted fabrics. A comparable approach has only been documented in the doctoral thesis of Cathryn Anneka Hall on the recycling of wool/polyacrylonitrile blends. Hall describes a company based in Prato,

Italy, that sorts knitted textiles according to coarse and fine knit structures to improve the quality of recycled fibres obtained from the recycling process (Hall, 2021).

2. Materials and Methods

2.1 Classification and Selection of Textile Waste Streams

Textile waste can generally be classified into post-industrial, pre-consumer, and post-consumer waste streams. Post-industrial textile waste consists of fibres, yarns, fabrics, and production residues generated during textile manufacturing processes (Baloyi et al., 2024). Pre-consumer textile waste refers to unused textile products and materials discarded after production but before reaching the end consumer, including overproduction, unsold inventory, and returned goods (Jaczynska & Mehta, 2025). Due to their known material compositions and controlled origins, both waste streams are generally well suited for mechanical recycling.

In contrast, PCTW comprises textiles discarded by consumers after use. It is characterized by highly variable fibre compositions, textile constructions, finishes, and levels of wear. This heterogeneity complicates sorting and recycling processes, making it difficult to establish consistent processing parameters and produce reproducible recycled fibres. Consequently, PCTW remains one of the major challenges for high-quality fibre-to-fibre recycling and highlights the need for advanced sorting strategies capable of generating well defined input streams for recycling processes (Thomas et al., 2024). As a result, fibres recovered from PCTW are predominantly downcycled into filling and insulation materials rather than being reintroduced into textile production. (Baloyi et al., 2024; Thomas et al., 2024; Wojnowska-Baryła et al., 2024).

This study focuses on PCTW as the most challenging waste stream for mechanical recycling. The selected materials were used to investigate process-related factors influencing recycled fibre quality and to evaluate potential measures for improving fibre-to-fibre recycling outcomes.

For the experimental study, two comparison bales of the article Acrylic Mix from the used clothing sorting plant SOEX were analyzed (see fig.1).



Figure 1: bale of the post-consumer textile waste article " Acrylic mix", sorted by the sorting company SOEX in Bitterfeld-Wolfen (Germany).

According to SOEX, Acrylic Mix is primarily used for the production of torn fibres, a category defined as fibres obtained from tearing of fabrics, knitwear, nonwovens, and finished textiles (Maetschke,

1978). Although Acrylic Mix consists mainly of knitted textiles, it contains a broad variety of fibre types in both mono-material and multi-material compositions. All woven textiles were removed from the experiment to avoid interference with the fibre separation behaviour specific to knits.

2.2 Differentiated Sorting Approach

Sorting for the experimental study was conducted in three stages:

1. Coarse sorting using the material recognition system from SOEX, based on major fibre categories
2. Fine sorting using existing labels within the pre-sorted groups
3. Sorting according to stitch density

This three-stage method enables more homogeneous material inputs for subsequent recycling steps and ensures comparability across samples. Sorting was conducted in March in Germany; therefore, seasonal and regional variations in PCTW, such as differences in summer/winter garment ratios and fibre quality, should be taken into account. Consequently, the wool fraction may vary in other contexts.

2.2.1 Coarse Sorting Using the Material Recognition System

The first sorting stage enable a coarse classification into plant- and animal based natural fibres, as well as cellulose-, and petroleum-based synthetic fibres. Sorting is carried out using SOEX's material recognition system, based on near-infrared spectroscopy and an RGB sensor. Using artificial intelligence (AI), individual garments are scanned and sorted according to material composition and color (*Sorting for Recycling*, n.d.). The predominant multi-fibre blends compositions pose a challenge for the material recognition system, as it can currently only detect blends of two different fibre types. Due to their differing fibre lengths and properties, all cellulose based fibres (cotton, linen, viscose, lyocell, modal, etc.) were removed. All animal-based fibres, both as mono-material compositions and within blends of petroleum-based synthetic fibres, as well as petroleum-based synthetic fibres served as the raw material for the planned experimental study. Figure 2 shows the total quantities of both PCTW bales resulting from the coarse sorting process, of which approximately 48% of the total mass is used as raw material for the test series carried out.

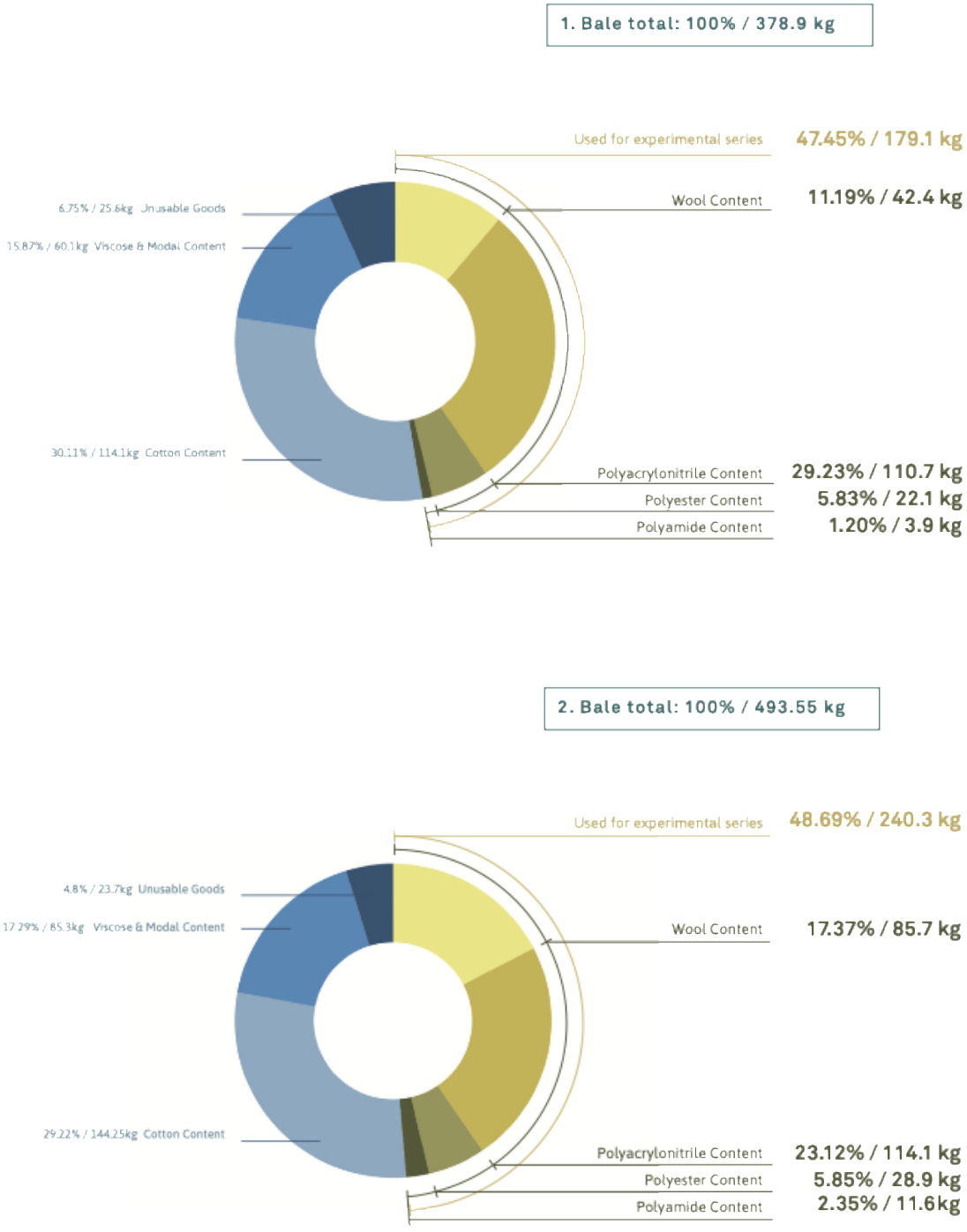


Figure 2: Evaluation of the 1st and 2nd Bale of Acrylic Mix.

2.2.2 Fine Sorting Using Existing Labels

Fine sorting is performed within the four predefined categories used for the experimental series and further differentiates the materials according to fibre composition and the proportion of individual fibres within each blend. Figure 3 summarizes the detailed fibre types per garment. In total, 1615

garments with a combined weight of 413.36 kg were individually examined (including sweaters, cardigans, socks, hats, scarves). Only 75 garments were made from 100% wool (WO), corresponding to 11.14 kg or 1.27% of the total PCTW sorted. Among the monomaterials, polyacrylonitrile (PAN) accounted for the largest share with 240 items and 72.51 kg, corresponding to 8.03% of the analyzed PCTW. Polyester (PES) amounted to 7.70 kg (0.88%), which was below the share of wool; alpaca was represented at 0,61kg (0.07%) and cashmere at 1,46 (0.09%). Polyamide did not occur as a monomaterial in the investigated knitted goods.

The majority of the material consisted of fibre blends. Non-recyclable fractions, including woven fabrics, jersey materials, metallized yarns, and fully applied sequins amounted 15,97kg (1,83%). A total share of 119,55 kg (13.70%) could not be considered in this analysis due to missing labels.

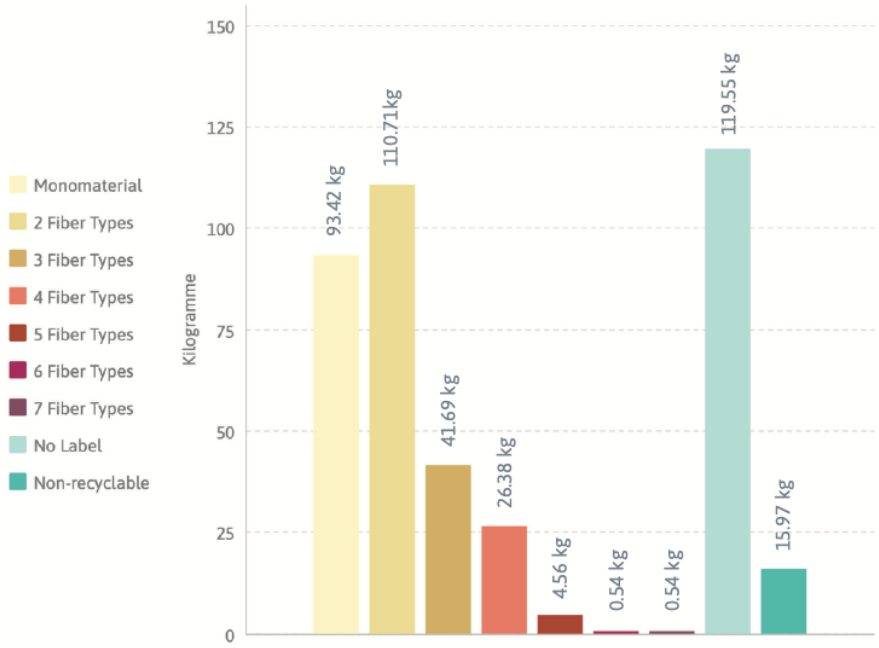


Figure 3: Analysis of the fine-sorting from the 1st and 2nd bale Acrylic Mix highlighting the variability in fibre types among individual garments.

This sorting step was conducted to assess the prevalence of mono-material textiles relative to fibre blends. The results further highlight the growing complexity of fibre compositions in current PCTW.

2.2.3 Sorting by Stitch Density

In the final sorting stage, PCTW is categorized according to stitch density to determine its influence on fibre separation.

Stitch density is assessed according to DIN EN 14971 (DIN, 2006), based on stitch count per length and area.

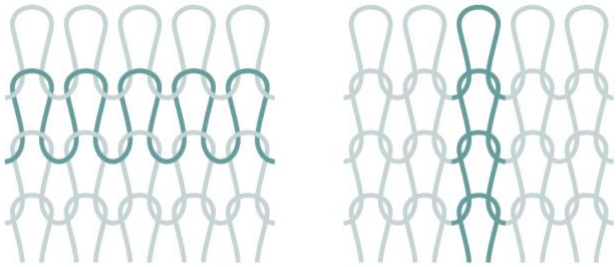


Figure 4: Schematic representation: course (left), wale (right).

Knitted fabrics are visually classified into coarse, medium, and fine stitch structures. To capture the inherent variability within each category, measurements are conducted on five reference knitted fabrics per group. Stitch density (stitches/decimeter²) was calculated as:

SD = SCD × SWD (stitch density = stitch course density x stitch wale density)

A single knit stitch connects to eight surrounding stitches, including four head bindings and four foot bindings (see fig. 5). The binding point is the contact point between two thread sections during stitch formation, with the head bindings in contact with the stitches from the row above and the foot bindings in contact with the stitches from the row below.

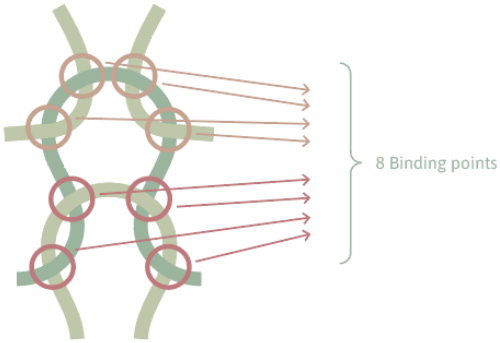


Figure 5: Binding points within a single knit stitch.

Coarse knits, with an average of 453 stitch counts and approx. 3624 binding points per 10×10 cm, are substantially easier to disassemble than fine knits, which exhibit 5133 stitch counts and approx. 41064 binding points per 10×10 cm. Stitch count and binding points affect fibre separation, fibre length reduction and nep formation during mechanical recycling.

The evaluation of stitch densities and calculated binding points within each category is shown in Figure 6.

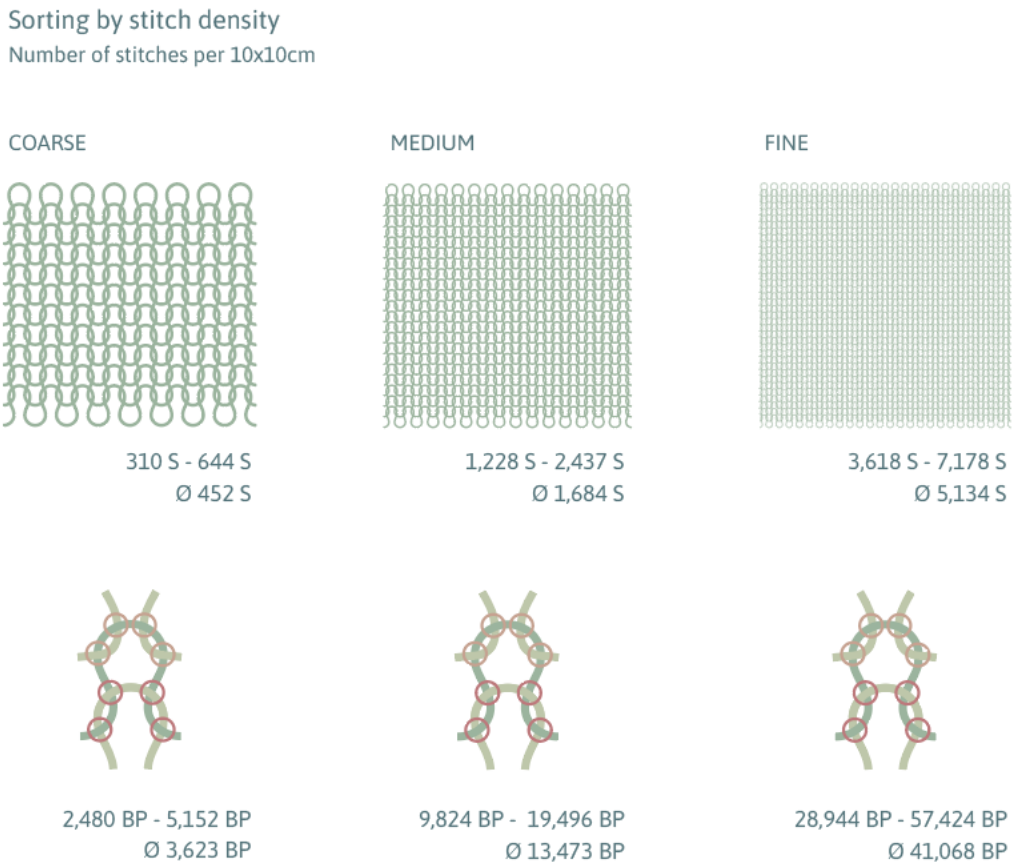


Figure 6: Evaluation of the stitch densities within the categories: coarse, medium, fine and the calculation of the total binding points (S = stitch, BP = binding point).

2.3 Mechanical Recycling

Starting with the pre-sorted PCTW, the mechanical recycling stages of cutting, tearing, carding, and spinning were carried out. The experimental focus was placed on the cutting and tearing stages, while the card slivers (continuous, slightly twisted strands of fibres produced during the carding process) and the resulting yarns primarily served as evaluation materials for assessing the effects of process variations on nep occurrence and yarn evenness. Figure 7 illustrates the principal processing steps of the mechanical recycling route.



Figure 7: Processing of the PCTW: a) cutting; b) tearing; c) carding; d) spinning.

A total of eighteen batches were produced for this purpose, consisting of eight batches 70% recycled wool / 30% primary wool and ten batches 70% recycled polyacrylonitrile / 30% primary wool. The addition of primary fibres was necessary to ensure sufficient fibre-to-fibre adhesion for subsequent yarn production despite the shortened recycled fibres.

To evaluate the influence of the opening processes on recycled fibre quality, different cut sizes and tearing intensities (none, one, or two tearing passes) were investigated. Particular attention was given to the cutting and tearing stages, as both processes were identified as potential sources of nep formation and therefore play a decisive role in the quality of the recycled fibres and the resulting yarns.

2.3.1 Cutting process

For comparison, PCTW consisting of 100% WO in two stitch density categories (coarse, fine) and 100% PAN in three stitch density categories (coarse, medium, fine) was cut into defined piece sizes (see Table 1). Due to the relatively low proportion of pure wool in the sorted material, a medium stitch density category could not be established for wool.

Cutting was performed using a PIERRET R45 guillotine cutter (Belgium) with a working width of 45 cm. The machine allows adjustable cut lengths ranging from 6 to 125 mm by varying the feed speed while maintaining a constant cutting frequency. A schematic representation of the cutting process is shown in Figure 8.

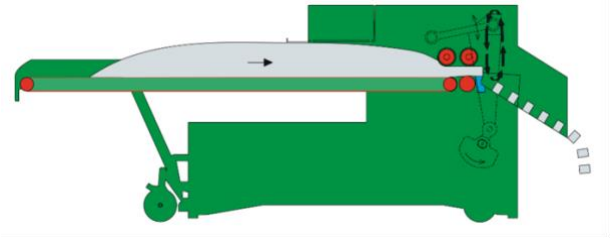


Figure 8: Schematic illustration of the cutting process using a PIERRET Industries R45 guillotine cutting machine (Pierret Industries, n.d.).

As shown in Figure 9-1a and 2a, the cutting edge was preferably oriented perpendicular to the yarn direction. First, the garments were cut into uniformly sized strips (Fig. 9b) and subsequently, in a second cutting step, into rectangular pieces at a 90° angle (Fig. 9c). In alignment with industrial practice, unsorted multi-fibre blends were additionally cut twice, regardless of yarn orientation (Fig. 9.3a). These blends were not separated by fibre types or stitch density, as they served as a comparative variant reflecting industrial handling.



Figure 9: 1a and 2a: material orientation in yarn direction; 1b, 2b and 3b: material after the first cut; 1c, 2c and 3c: material after the second cut; 3a: material orientation independent of yarn direction

In particular, blends containing elastane exhibited significant deficiencies in cutting quality. These included fused cut edges, perforated areas that were not completely severed (Fig. 9-3c), diagonally oriented yarn alignments (Fig. 9-3b), and strongly varying cut sizes.

The direct comparison between 100% WO and 100% PAN revealed clear material-dependent differences. In WO textiles, precise cutting results could generally be achieved (Fig. 10).



Figure 10: Wool batches with precise cutting results.

In contrast, the cutting results for polyacrylonitrile were consistently inadequate. It was not possible to produce single-layer, clearly defined cut edges. Instead, multiple layers were regularly fused together at the cut edges, resulting in a cushion-like appearance. Even with a single-layer material feed (Fig. 9-2a), this phenomenon could not be reduced. It rather led to material accumulation in the feeding system (Fig. 11).



Figure 11: Fused cut edges of PAN knitted fabrics.

Maetschke already describes in this context that petroleum-based synthetic fibres tend to form melt neps during the opening process due to high thermal stress, and therefore cannot be adequately opened (Maetschke, 1978). To analyze the effects of these fusions, two additional batches were manually cut to the desired sizes using standard tailor's scissors (see Table 1). No fused cut edges were observed in these samples.

The observed fused cut edges in PAN were analyzed using a scanning electron microscope (SEM). The images are shown in Figure 12. The images reveal distinct thermally melted regions at the fibre cut edges, where surrounding fibres and fibre layers have partially fused together during the cutting process. These fused areas form localized agglomerations that differ significantly from the original fibre structure.

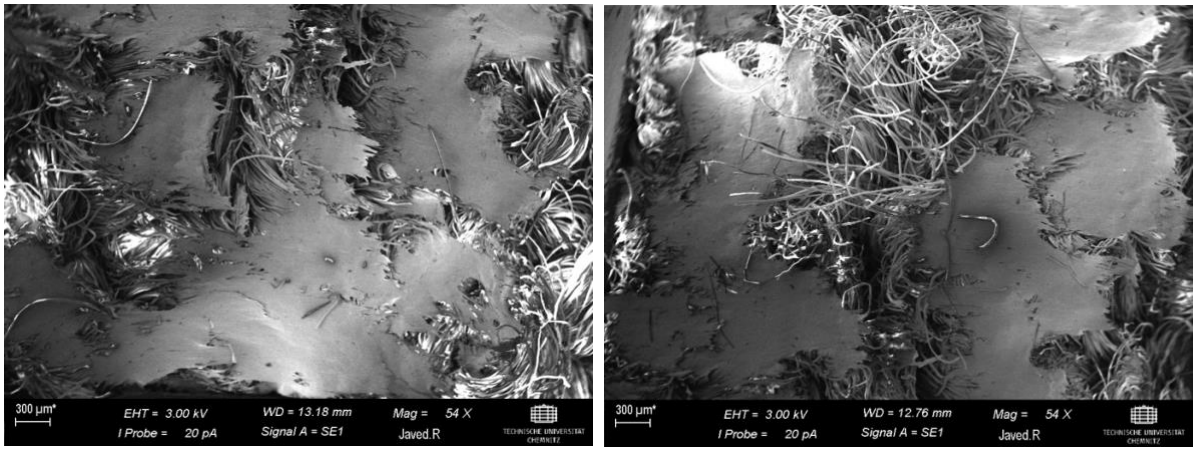


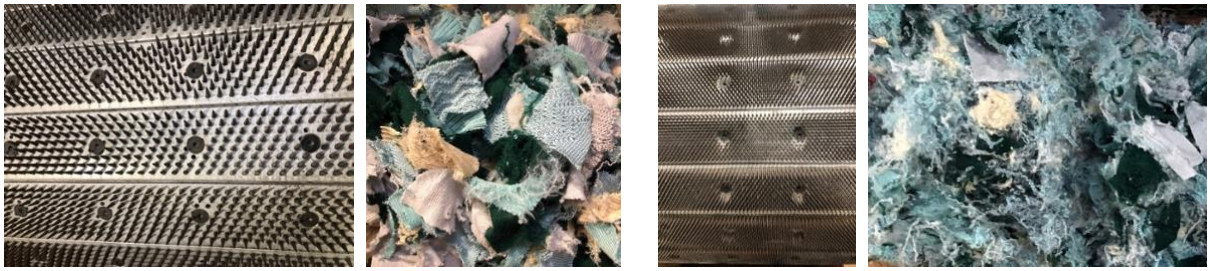
Figure 12: SEM micrographs of fused polyacrylic cut edges.

2.3.2 Tearing process

To exclude additional variables, all tearing trials were conducted using a standardized cut size of 5 cm² and constant machine settings. During tearing, the textile pieces are mechanically opened by rotating pinned drums that progressively separate the fabric structure into individual fibres and fibre bundles.

Three process variants were investigated: (i) no tearing, (ii) one tearing pass using an SS 8 pinned drum, and (iii) two tearing passes using SS 8 and SG 16 pinned drums (see fig. 13 and table 1). The SS

8 drum is equipped with a comparatively coarse pin configuration and therefore provides a more intensive opening action, whereas the SG 16 drum features a finer pin configuration for gentler fibre separation and further material opening. This approach enabled a systematic comparison of different tearing intensities and their effects on fibre quality.



a) *b)* *c)* *d)*

Figure 13: Pin drums used in the tearing process: a) SS 8 pin drum; b)torn PCTW after the 1st tearing process; c) SG16 pin drum; d) torn PCTW after the 2nd tearing process.

The tearing trials were performed using an AUTEFA SERVO 60 laboratory tearing unit equipped with interchangeable pinned drums representative of industrial practice.

Table 1: Listing of cut sizes and tearing passes within the defined batches.

Sample no.	Fibre material	Defined stitch density	Yarn direction	Cutting size (mm)	Tearing passes
1	PCTW 100% Wool	coarse	✓	50 x 50	1x SS 8
2	PCTW 100% Wool	coarse	✓	50 x 50	1x SS 8 1x SG 16
3	PCTW 100% Wool	medium	✓	50 x 50	1x SS 8
4	PCTW 100% Wool	fine	✓	50 x 50	1x SS 8 1x SG 16
5	PCTW 100% Wool	coarse	✓	20 x 40	No tearing
6	PCTW 100% Wool	fine	✓	20 x 40	No tearing
7	PCTW 100% Wool	coarse	✓	13 x 30	No tearing
8	PCTW 100% Wool	coarse	✓	6 x 20	No tearing
9	PCTW 100% PAN	coarse	✓	50 x 50	1x SS 8
10	PCTW 100% PAN	medium	✓	50 x 50	1x SS 8
11	PCTW 100% PAN	medium	✓	50 x 50	1x SS 8 1x SG 16
12	PCTW 100% PAN	medium	✓	20 x 40 Cut manually	No tearing
13	PCTW 100% PAN	fine	✓	50 x 50	1x SS 8
14	PCTW 100% PAN	fine	✓	50 x 50	1x SS 8 1x SG 16
15	PCTW 100% PAN	medium	✓	30 x 50 Cut manually	No tearing
16	PCTW 100% PAN	coarse	✓	20 x 40	No tearing
17	PCTW 100% PAN	fine-coarse	✗	13 x 30	No tearing
18	PCTW 100% PAN	fine-coarse	✗	20 x 40	No tearing

2.4 Nep Evaluation of Card Slivers

The evaluation of neps in the card slivers was conducted using the MDTA 4 instrument (Textechno). The resulting sliver and the separated trash were subsequently analyzed using the CLIF system (The Textechno Group, n.d.), which comprises a transmitted light scanner, a high-performance PC with GPU, and dedicated CLIF software. A total of five measurements were performed for each card sliver batch using the MDTA 4.

For sample preparation, 0.5 g of card sliver per measurement was evenly distributed in thin layers onto the conveyor belt of the MDTA 4. During processing, an opening roller separates neps and yarn fragments, which are collected in a trash container, while the remaining fine fibre material is deposited as a sliver in a drum. In a first step, the fibre sliver is divided into five equal parts and placed vertically onto the CLIF scanner. The first partial measurement is then carried out, detecting neps in the size ranges of 0.5–0.7 mm, 0.7–1.0 mm, and > 1.0 mm, as well as yarn fragments.

For the second partial measurement, the contents of the trash container are subsequently placed on the scanner, analyzed for neps and fibre fragments, and the results are added to those of the first measurement. Figure 14 presents the evaluation of the 8 batches 70%rWO/30%WO and figure 15 the 10 batches 70%rPAN/30%WO.

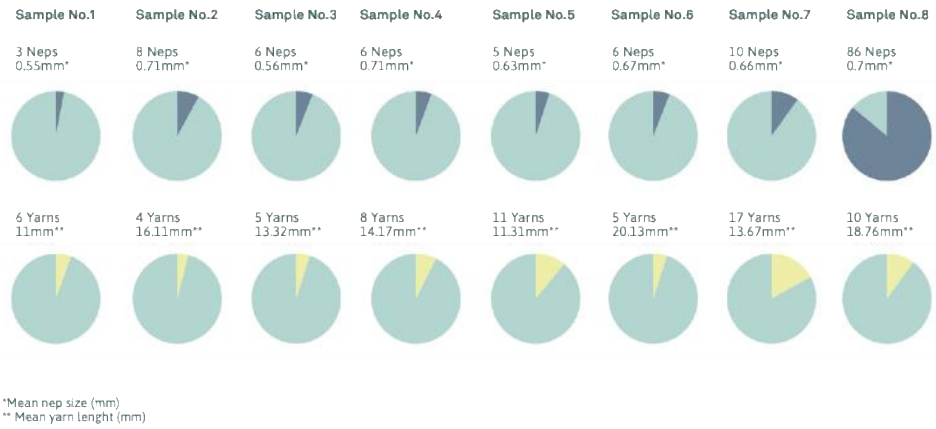


Figure 14: Nep and Yarn Analysis of the batches using an average of 0,5g sliver (70%rWo/30%Wo).

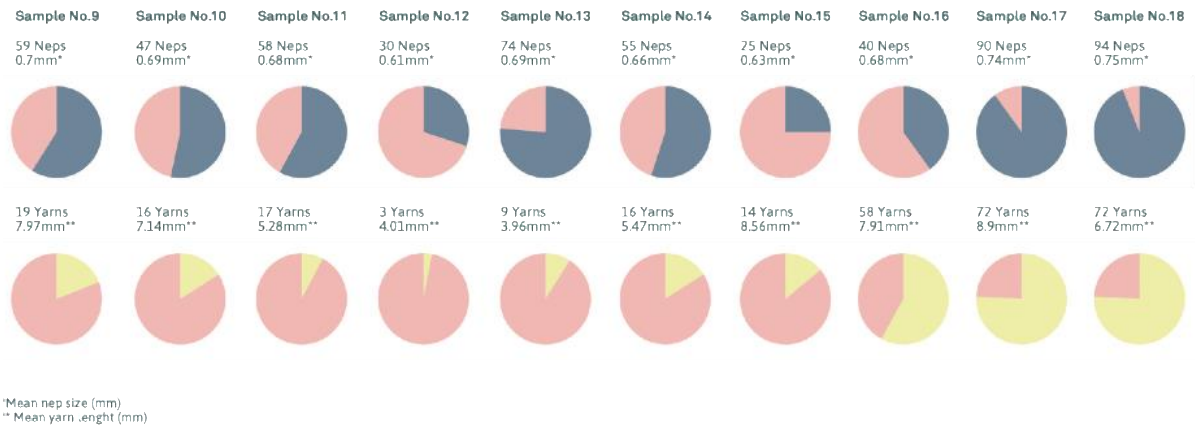


Figure 15: Nep and Yarn Analysis of the batches using an average of 0,5g sliver (70%rPAN/30%Wo).

2.5 Yarn Quality Evaluation

To evaluate the effect of neps and yarn fragments present in the card sliver on the properties of the final rotor yarn, the produced rotor yarns were tested using the Evenness Tester by Mesdan. Due to insufficient fibre lengths, rotor yarn could not be produced from card slivers with a composition of 70% rWO/30% WO. Consequently, evenness measurements were only carried out on batches consisting of 70% rPAN/30% WO.

Prior to evenness testing, the yarn count of each sample was determined. For this purpose, a six-arm precision wrap reel (Mesdan, Model 161M, Serial No. 2466/2019) with a circumference of (1000 ± 2.5) mm was used. For each measurement, a 10 m yarn length was wound and weighed individually using a balance with a precision of three decimal places. The yarn count was then calculated from the measured length and mass according to the following Equation:

$$\frac{l(m)}{m(g)}$$

l = yarn length (10 m)
m = yarn mass (g)

Due to the heterogeneous fibre blends obtained from PCTW, the target nominal yarn count could not always be achieved. Instead, considerable variations in yarn fineness were observed, in some cases even within a single yarn package. Since the Mesdan Evenness Tester requires the nominal yarn count to be defined before each measurement, a single measurement setting could not be applied to all samples. Consequently, the measurement settings were adjusted according to the previously determined yarn count, and for some yarn samples, multiple evenness measurements were performed using different predefined nominal yarn count settings. It is evident that the selected measurement setting considerably influences the measured proportions of thin and thick places. Therefore, the obtained values should be interpreted on a relative rather than an absolute basis. Depending on the intended application, these pronounced variations may be of particular relevance. For each batch, 300 m of yarn were measured. The evaluation of yarn samples 9-18 is presented in Figure16.

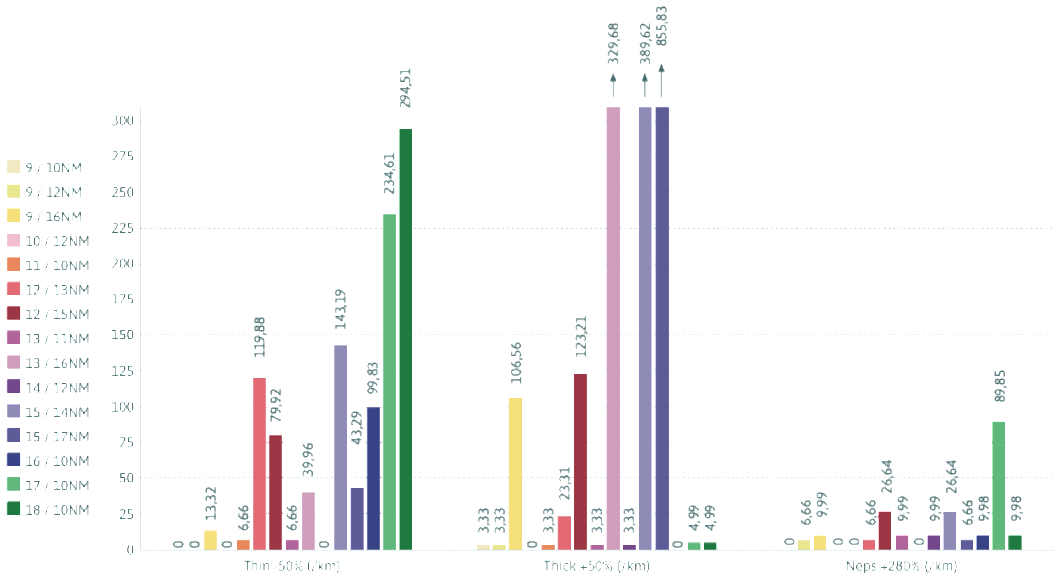


Figure 16: Evenness Evaluation of yarn batches containing 70%rPAN/30%Wo.

3 Evaluation of Findings

The nep evaluation reveals substantial differences between the rWO/WO samples (No. 1-8) and the rPAN/WO samples (No. 9-18). This particularly indicates a direct influence of melt surfaces generated during the cutting process of PAN knitted fabrics. Another contributing factor to increased nep formation is the cut size, which is especially evident in sample No. 8. Despite consisting of pure wool, this sample exhibits a higher nep content, as the cutting height of 6 mm resulted in a large proportion of short fibres in the card sliver.

In samples No. 12 and 15 from the rPAN/WO blend, a comparatively low nep content can be observed. In comparison with Table 1, it becomes evident that these samples were cut manually, thereby preventing the formation of melt-induced defects during cutting. Notably, samples No. 17 and 18 exhibit the highest values for both neps and yarn remnants. This can be attributed to unsorted material that had not been previously classified by fibre composition or knitting structure. Furthermore, materials processed without tearing generally showed the lowest nep content. These findings highlight the importance of proper sorting as well as the influence of cutting and tearing conditions in achieving a more homogeneous card sliver and reducing both nep formation and residual yarn content.

Sample 17 exhibited the highest nep content. Although it showed similar numbers of thick and thin places compared to Sample 18, it differed considerably in terms of nep occurrence, which can primarily be attributed to its smaller cut size. Samples 12 and 15 also displayed comparatively high levels of neps as well as thick and thin places. Both samples were cut manually and originated from fabrics with a medium stitch density, contrasting with the results obtained from the card sliver analysis. In addition, Samples 13 and 14 showed elevated nep contents, which may be related to the fine stitch density of the knitted fabrics.

Overall, it can be concluded that nep occurrence can be more reliably detected at the card sliver stage, as a consistent amount of material is analyzed. During subsequent processing into rotor yarn, it is possible that neps are either removed or further broken down during fibre re-opening. However, due to the irregular yarn fineness, directly comparable values cannot be obtained at the yarn stage. This limitation may potentially be mitigated by achieving a more homogeneous card sliver and by optimizing machine parameters.

4 Conclusion

The presented three-stage sorting method has the potential to be fully automated in the future, including AI-based detection of stitch density. However, the currently limited industrial demand for finely sorted PCTW hinders the large-scale implementation of such systems. A broader market uptake of high-quality recycled fibres would be necessary to justify further technological development and refinement.

The conducted analyses of nep formation and the resulting yarn irregularities clearly demonstrate the direct impact of more precisely sorted PCTW. Consequently, greater attention must be given to sorting processes, alongside the development of advanced sorting systems capable of producing more homogeneous fibre feedstock in terms of both fibre composition and stitch density. Furthermore, the results indicate that the cutting and tearing stages significantly influence recycled fibre quality. In particular, thermally melted cut edges of polyacrylic-containing materials were identified as a

potential source of nep formation, highlighting the importance of process optimization in addition to material sorting.

In particular, the evenness tests indicate that existing measurement methods in the textile sector are not adequately designed for recycled materials. In contrast to the MDTA 4 device, which specifically addresses challenges associated with recycling processes, conventional evenness measurements appear to be of limited relevance for recycled yarns. This is primarily due to the inherent difficulty in achieving uniformity in recycled yarns, especially when processing PCTW. Consequently, the development of adapted or entirely new measurement parameters may be required to appropriately assess the quality of such materials.

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About the Authors:

Magdalena Kohler: She works at the intersection of design, technology, and sustainability. As a PhD candidate at Chemnitz University of Technology and research assistant in the EU funded project STELEC, she investigates textile circularity, wool lifecycles, and the transformation of textile waste into new material opportunities.

Holger Cebulla: Professor of Textile Technologies at Chemnitz University of Technology, specializing in textile engineering, sustainable textile production and textile machinery. He combines academic research with international industrial leadership and has founded several initiatives promoting sustainability and innovation in the textile sector.

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